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[5352]-177

S.E. (Information Technology) (II Sem.) EXAMINATION, 2018
PROCESSOR ARCHITECTURE AND INTERFACING
(2012 PATTERN)

Time : Two Hours

Maximum Marks : 50

- N.B. :—** (i) Answer Q. No. 1 or Q. No. 2, Q. No. 3 or Q. No. 4, Q. No. 5 or Q. No. 6, Q. No. 7 or Q. No. 8.
(ii) Neat diagrams must be drawn wherever necessary.
(iii) Figures to the right indicate full marks.
(iv) Assume suitable data if necessary.

1. (a) What is the purpose of assembler directives ? Explain the following assembler directives used in 80386 programming :
(i) DW
(ii) MACRO
(iii) EXTRN [6]
(b) Differentiate between the steps performed by processor when near and far procedure call occurs. [6]

Or

2. (a) Explain how pipelined processing improves performance of the system as compared to non-pipelined processing of 80386. [6]
(b) Draw and explain EFlag register of 80386 processor. [6]

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3. (a) Explain the paging mechanism implemented in 80386 and how the protection features are achieved. [6]
- (b) What is Multitasking ? How is it achieved in 80386 processor ? [6]

Or

4. (a) Explain the working of confirming code segment in 80386 processor. [6]
- (b) What is dual core processor ? Give the differences between single core and dual core processor. [6]
5. (a) Explain function of the following pins of 8051 micro-controller : [6]
- (i) ALE
- (ii) INT0
- (b) Draw and explain memory organisation of 8051 micro-controller. [7]

Or

6. (a) Explain the following instructions of 8051 microcontroller : [6]
- (i) ANL C, bit
- (ii) RRC A
- (iii) MOVX A, @DPTR
- (b) Write a program to continuously send out to port 0 of 8051 the alternating values of 55H and AAH. [7]

7. (a) Explain PCON and TCON registers of 8051 microcontroller. [6]
- (b) State and explain SFR's related to serial communication in 8051 microcontroller. [7]

Or

8. (a) Draw and explain the Mode 0 and Mode 1 of timer in 8051 microcontroller. [7]
- (b) Explain the interrupt programming with the help of IE and IP SFR in 8051 microcontroller. [6]